
From Score-Based Foundations to Latent Diffusion: A Journey into Multimodal Image Synthesis

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Abstract

This project presents a PyTorch implementation of diffusion-based generative models, progressing through 4 phases: (1) denoising score matching with Langevin dynamics on 2D toy datasets; (2) Noise-Conditional Score Networks (NCSN) with annealed multi-scale noise schedules; (3) Denoising Diffusion Probabilistic Models (DDPM) scaled to MNIST via a time-conditioned U-Net stabilized with GroupNorm, EMA and residual depth; and (4) Latent Diffusion Models (LDM) built on a custom VAE regularized by 3 losses. Text-to-image synthesis is enabled by a frozen CLIP text encoder, 2D cross-attention and classifier-free guidance with DDIM sampling. The framework is validated on MNIST, Oxford Flowers-102 and 2 datasets collected with my smartphone and curated through CLIP-based semantic deduplication.

1. Overview

The implementation progresses incrementally from denoising score matching on 2D toy data, through NCSN and DDPM on MNIST, up to a latent diffusion model on customized datasets and a final text-to-image LDM with CLIP conditioning and classifier-free guidance. Table 1 summarizes the architecture, parameter count, model size and dataset of every network in the project.

2. Score-Based Models on 2D Toy Data

The first dataset consists of 10.000 points sampled uniformly on a circle via polar coordinates, served through a `TensorDataset/DataLoader` pipeline. `ScoreMLPPoints`, a small MLP with 2 hidden layers of width 20 receives a noisy point (x, y) and predicts the

injected Gaussian noise in 15 epochs. Training follows denoising score matching: each batch is perturbed as $\tilde{\mathbf{x}} = \mathbf{x} + \sigma\epsilon$ and I minimized the MSE between predicted and true noise, using Adam optimizer with $lr = 10^{-3}$. Since $\nabla_{\tilde{\mathbf{x}}} \log q_{\sigma}(\tilde{\mathbf{x}} | \mathbf{x}) = -\epsilon/\sigma$, the score is recovered from the predicted noise and samples are generated from scaled Gaussian noise via Langevin dynamics:

$$\tilde{\mathbf{x}}^k = \tilde{\mathbf{x}}^{k-1} + \frac{\lambda}{2} \mathbf{s}_{\theta}(\tilde{\mathbf{x}}^{k-1}, \sigma) + \sqrt{\lambda} \mathbf{z}^k, \quad \mathbf{z}^k \sim \mathcal{N}(0, I),$$

with intermediate steps plotted in Plotly grids to visualize convergence onto the circle. Among five activations trained identically (ReLU, LeakyReLU, ELU, Sigmoid, Tanh), ReLU gave the best sample quality in less training time. Generalization was verified on a sinusoid dataset, including loss curves and selecting the noise level σ from the set $\{0.1, 0.3, 0.8\}$.

3. Noise-Conditional Score Network (NCSN)

Since score estimation is unreliable in low-density regions, I defined `CondMLP`, a MLP which conditions on the noise level by adding a linear embedding of σ to the coordinate embedding. A geometric schedule of $L = 8$ scales from $\sigma_{\max} = 4.0$ to $\sigma_{\min} = 0.05$ is sampled randomly during training and sampling uses annealed Langevin dynamics (125 steps per scale, step size $\epsilon(\sigma_i/\sigma_L)^2$). This helped the model learn to cover a much wider range of the data.

4. DDPM: 2D Data and MNIST Digit Generation

DDPM defines a Markov chain in $T = 1000$ steps with a linear schedule $\beta_t \in [10^{-4}, 0.02]$; defining $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_s \alpha_s$, the reparametrization $x^{(t)} = \sqrt{\bar{\alpha}_t} x^{(0)} + \sqrt{1 - \bar{\alpha}_t} \epsilon$ reaches any timestep in $O(1)$ without the need to pass through each step of the chain. Timesteps are encoded with sinusoidal position embeddings passed through a small MLP. A time-conditioned MLP first reproduced the sinusoid dataset; the procedure was then scaled to MNIST with a compact U-Net (conv stages 32/64/128, transposed-conv decoder, time embeddings added to each layer). I computed and registered activation statistics in the bottle-

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neck layer of the network via a forward hook. This motivated 3 fixes: **(i) GroupNorm** with 8 groups shrank activation ranges from roughly $(-13, 8)$ to $(-5, 4]$; **(ii) EMA** of the weights (decay 0.999) at sampling time (EMA and GroupNorm increased the stability and result quality); **(iii)** a deeper residual U-Net (UNetDeepV2) with UBlocks made of 2 convolutions and concatenation skips. Trained for 50 epochs at $lr = 2 \times 10^{-4}$, it produces recognizable digits from pure noise.

5. Data Collection and Preprocessing

Two original datasets were captured with an iPhone 13 Pro: a *botanical garden* set (Orto Botanico di Roma, Hanami Festival, April 2026; 4K 30 fps video) and a *museum* set (Galleria Nazionale d’Arte Moderna e Contemporanea, July 2026; HD 30 fps). Videos were decomposed into frames with OpenCV and resized. Since consecutive frames are redundant, I implemented **semantic deduplication** encoding each frame with CLIP ViT-B/32 and discarding it when cosine similarity with the last kept embedding (since they’re sequential) exceeds 0.95, reducing the botanical set from $\sim 74k$ to 14,250 frames and the museum set from 38,763 to 4,690. I also used a frozen CLIP model to label the botanical set with per-image captions. To remove disk-I/O and encoder bottlenecks, scaled VAE latents and caption embeddings are cached in RAM before training; a configuration block selects dataset and hyperparameters and every stage saves and resumes from checkpoints.

6. Latent Diffusion on Custom Datasets

I implemented a convolutional VAE ($\sim 5.8M$ parameters; SiLU ResBlocks, three down/upsampling stages, 4 latent channels) compresses 256×128 images into the latent space with loss $\mathcal{L} = \mathcal{L}_{L1} + \omega_p \mathcal{L}_{LPIPS} + \omega_{kl} \mathcal{L}_{KL}$ ($\omega_{kl} = 10^{-4}$): L1 preserves sharp edges, the frozen VGG-based LPIPS term enforces perceptual similarity (more robust than pixel differences, since it considers high-level features extracted by VGG) and KL regularizes the latent distribution. Training uses mixed precision and I computed a latent **scale factor** $s = 1/\sigma_z$ to normalize latents to unit variance and stabilize training. Reconstructions on the museum set after 100 epochs are shown in Figure 4.

The diffusion prior is a LatentUNet (base 128, time dim. 512) with residual blocks and multi-head *self-attention* at lower resolutions, trained with a **cosine noise schedule**, the **v-prediction** objective $v = \sqrt{\alpha_t} \epsilon - \sqrt{1 - \alpha_t} x_0$ and EMA (decay 0.9999). I implemented both classical **DDPM** and step-skipping **DDIM** (e.g. 50 steps) samplers. To check quality I used **Fréchet Inception Distance** (based on Inception-v3, 2048-d features; comparing the distributions of 200 real vs. 200 generated images). A progressive-

sampling function decodes intermediate latents to visualize how the noise decoded by VAE incrementally takes the shape of an image (Figure 5).

7. Multimodality: Text-to-Image Generation

Text conditioning uses a frozen pre-trained **CLIP text encoder** (ViT-B/32): prompts are tokenized to 77 tokens and mapped to hidden-state embeddings, cached for efficiency. On **Oxford Flowers-102** complete dataset class labels become prompts (‘a photo of a {*flower name*}’). A CrossAttentionBlock lets image features (queries) attend over text embeddings (keys/values) and the MultiModalLatentUNet interleaves residual, self-attention and cross-attention blocks at every scale, with a zero-initialized output layer. I implemented classifier-free guidance by randomly omitting 10% of text prompts during training to improve conditional fidelity, and utilized gradient clipping to stabilize training by bounding the norm of the gradients. I optimized through AdamW with weight decay and a cosine-annealed LR over 350 epochs using EMA. Sampling is defined with CFG mixing $\tilde{\epsilon} = \epsilon_{\emptyset} + w(\epsilon_c - \epsilon_{\emptyset})$, timestep respacing and latent x_0 clipping for stability at high guidance scales. The same procedure is finally applied to the captioned botanical dataset, generating new images from random dataset captions and free user prompts.

8. Results

On the 2D benchmarks, Langevin sampling recovers the circle and sinusoid manifolds, with ReLU the best activation and annealed NCSN sampling clearly improving coverage over the single-scale model. On MNIST, GroupNorm compresses bottleneck activations from roughly $(-13, 8)$ to $(-5, 4]$ and the EMA-weighted deeper U-Net yields recognizable digits after 50 epochs.

For the latent pipeline, the VAE diagnostic in Figure 1 shows that Flowers-102 reconstructions preserve color, pose and structure with only mild smoothing of high-frequency texture, confirming the 4-channel latent space retains enough information for diffusion. The Flowers-102 LDM also produces coherent prompt-conditioned samples, e.g. the pink-yellow dahlias in Figure 3. The text-conditioned LDM responds correctly to free prompts: for “a cactus plant” (Figure 2; 16 samples, CFG scale 4.0, 250 respaced steps, EMA weights) it generates varied 64x64 greenhouse sceneries with distinct cactus shapes consistent with the botanical training data. Diversity across the grid indicates there’s no mode collapse, although fine details remain soft because of the small size and low quality of the training data (this choice was due to computer memory and GPU limits).

Bibliography. (Vaswani et al., 2017; Ho et al., 2020; Wu & He, 2018; Rombach et al., 2022; Zhang et al., 2018; Ho & Salimans, 2022; Nichol & Dhariwal, 2021; Saharia et al., 2022; Radford et al., 2021; Pascanu et al., 2013; Heusel et al., 2017)

9. Statement on the use of AI

AI acted as assistants to help me learn the theory faster, write repetitive code smoothly and fix bugs when I got stuck. I made sure to thoroughly review everything so that I fully understand the underlying mechanics and code.

Here is exactly how I used each tool:

- **Gemini:** I used it to look up literature, brainstorm ideas and break down complex concepts, like how Latent Diffusion Models and Cross-Attention blocks work in industrial architectures.
- **Gemini Extended Flash:** I used this model specifically for debugging and code verification. Whenever I ran into weird runtime errors or pipeline bottlenecks in PyTorch, I used it to help me isolate the bugs and make sure my training loops were working correctly. I used it also to update the Github README, make the network architecture table and bibliography in this report (this one was a problem, I had to retrieve each bibtex personally because it messed up with some metadata such as the pages) and check grammar, spelling and content mistakes.
- **GitHub Copilot:** I used this directly in my code editor for real-time autocompletion. It was great for accelerating the workflow by writing repetitive boilerplate code, like standard data preprocessing scripts and basic dataset/dataloader setups.

While these tools definitely sped up my development process, every line of code was manually checked, tested and tweaked by me to fit the project’s needs.

References

Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., and Hochreiter, S. Gans trained by a two time-scale update rule converge to a local nash equilibrium. In Guyon, I., Luxburg, U. V., Bengio, S., Wallach, H., Fergus, R., Vishwanathan, S., and Garnett, R. (eds.), *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/file/8a1d694707eb0fefe65871369074926d-Paper.pdf.

Ho, J. and Salimans, T. Classifier-free diffusion guidance. *CoRR*, abs/2207.12598, 2022. doi: 10.48550/ARXIV.2207.12598. URL <https://doi.org/10.48550/arXiv.2207.12598>.

Ho, J., Jain, A., and Abbeel, P. Denoising diffusion probabilistic models. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H. (eds.), *Advances in Neural Information Processing Systems*, volume 33, pp. 6840–6851. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/file/4c5bcfec8584af0d967f1ab10179ca4b-Paper.pdf.

Nichol, A. Q. and Dhariwal, P. Improved denoising diffusion probabilistic models. In Meila, M. and Zhang, T. (eds.), *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pp. 8162–8171. PMLR, 18–24 Jul 2021. URL <https://proceedings.mlr.press/v139/nichol21a.html>.

Pascanu, R., Mikolov, T., and Bengio, Y. On the difficulty of training recurrent neural networks. In Dasgupta, S. and McAllester, D. (eds.), *Proceedings of the 30th International Conference on Machine Learning*, volume 28 of *Proceedings of Machine Learning Research*, pp. 1310–1318, Atlanta, Georgia, USA, 17–19 Jun 2013. PMLR. URL <https://proceedings.mlr.press/v28/pascanu13.html>.

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., and Sutskever, I. Learning transferable visual models from natural language supervision. In Meila, M. and Zhang, T. (eds.), *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pp. 8748–8763. PMLR, 18–24 Jul 2021. URL <https://proceedings.mlr.press/v139/radford21a.html>.

Rombach, R., Blattmann, A., Lorenz, D., Esser, P., and Ommer, B. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 10684–10695, June 2022.

Saharia, C., Chan, W., Saxena, S., Li, L., Whang, J., Denton, E. L., Ghasemipour, K., Gontijo Lopes, R., Karagol Ayan, B., Salimans, T., Ho, J., Fleet, D., and Norouzi, M. Photorealistic text-to-image diffusion models with deep language understanding. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A. (eds.), *Advances*

in *Neural Information Processing Systems*, volume 35, pp. 36479–36494. Curran Associates, Inc., 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/file/ec795aeadae0b7d230fa35cbaf04c041-Paper-Conference.pdf.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. Attention is all you need. In Guyon, I., Luxburg, U. V., Bengio, S., Wallach, H., Fergus, R., Vishwanathan, S., and Garnett, R. (eds.), *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf.

Wu, Y. and He, K. Group normalization. In Ferrari, V., Hebert, M., Sminchisescu, C., and Weiss, Y. (eds.), *Computer Vision - ECCV 2018 - 15th European Conference, Munich, Germany, September 8-14, 2018, Proceedings, Part XIII*, volume 11217 of *Lecture Notes in Computer Science*, pp. 3–19. Springer, 2018. doi: 10.1007/978-3-030-01261-8_1. URL https://doi.org/10.1007/978-3-030-01261-8_1.

Zhang, R., Isola, P., Efros, A. A., Shechtman, E., and Wang, O. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2018.

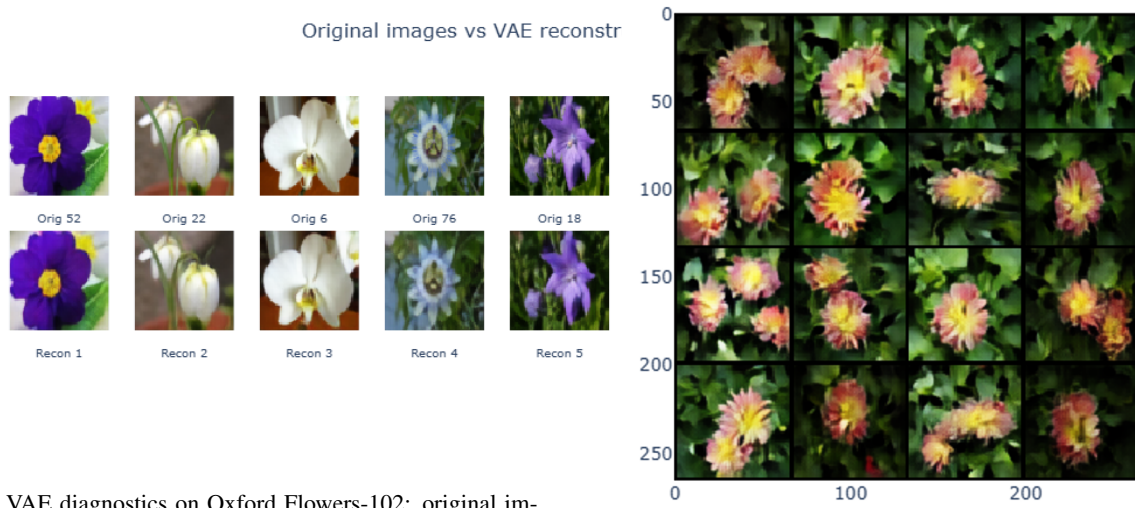


Figure 1. VAE diagnostics on Oxford Flowers-102: original images (top row) vs. their reconstructions (bottom row). Color, shape and composition are preserved, with slight smoothing of fine texture.

Figure 3. 16 pink-yellow dahlias generated through latent diffusion on Oxford Flowers-102 dataset

user prompt: 'a cactus plant'

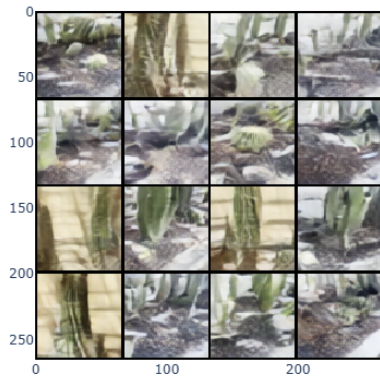


Figure 2. Text-to-image generation from the custom botanical LDM for the free-form user prompt "a cactus plant" (16 samples, CFG = 4.0, 250 sampling steps).



Figure 4. VAE reconstruction of museum images after 100 training epochs.

Table 1. Architectural Summary and Profiling of Evaluated Neural Networks

Network Name	Input Shape	# Params	Model Size (Params / Total)	Dataset
ScoreMLPPoints	$[B, 2]$	50,178	0.20 MB / 0.27 MB	2D Toy Data
CondMLPFirstV	$[B, 2] + [B, 1]$	50,434	0.20 MB / 0.28 MB	2D Toy Data
DPM	$[B, 16, 2]$	99,714	0.33 MB / 1.17 MB	2D Toy Data
UNet	$[B, 1, 28, 28]$	314,689	1.26 MB / 11.90 MB	MNIST
UNetPlusGroupNorm	$[B, 1, 28, 28]$	348,385	1.39 MB / 22.47 MB	MNIST
UNetPlusEmaAndNorm	$[B, 1, 28, 28]$	348,385	1.39 MB / 22.47 MB	MNIST
UNetDeepV2	$[B, 1, 28, 28]$	1,574,273	6.30 MB / 86.08 MB	MNIST
VAE	$[B, 3, 256, 128]$	1,487,755	5.95 MB / 4120.56 MB	Custom (Museum)
LatentUNet	$[B, 4, 32, 16]$	17,657,348	70.63 MB / 278.05 MB	Custom (Museum)
VAEMultiModal	$[B, 3, 64, 64]$	1,487,755	5.95 MB / 520.28 MB	Flowers-102 / Botanical
MultiModalLatentUnet	$[B, 4, 8, 8] + [B, 77, 512]$	4,492,356	17.97 MB / 81.10 MB	Flowers-102 / Botanical

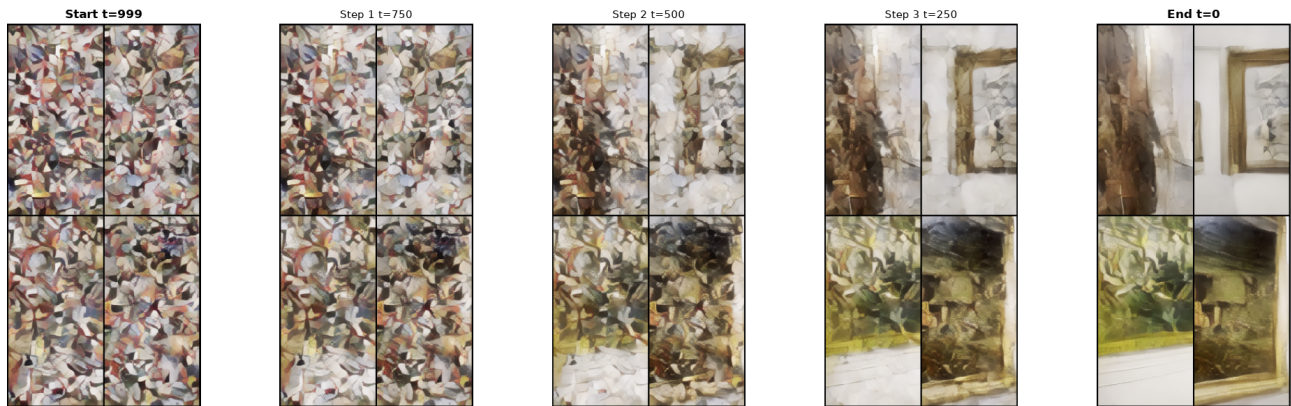


Figure 5. Museum images generated after 100 VAE epochs and 145 Latent Diffusion training epochs on an NVIDIA GeForce RTX 4070, achieving a FID score of 205.673. The mosaic effect at $t = 999$ appears because the VAE’s upscaling process forces isolated latent noise values to blend into correlated blocks of color in the pixel space. The slightly lower generation quality compared to the botanical results is due to the museum dataset’s high visual variability; learning a complex data manifold containing diverse elements like paintings, sculptures, and people heavily stretches the model’s capacity.